

Modeling and simulation of a hybrid power system for Hospital in Sabratha, Libya via HOMER software

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Abstract—With the rise in electricity prices and the need to reduce environmental pollution, renewable energy-based hybrid energy systems became the ideal solution to these problems. Therefore, New technologies are being developed to extract energy from every available source and store the excess energy generated for later usage. Thus, the hybrid energy system (HES) is a collection of various renewable power resources such as solar, wind, geothermal, hydro, tidal, etc. This paper studies the feasibility of powering sabratha's general hospital with electricity using a solar-wind hybrid power plant. The solar and wind characteristics of the site was taken from the National Aeronautics and Space Administration agency (NASA). A 10kW wind turbine was used for the wind system while a 350w solar panel was used in the solar system. Therefore, the optimizations and simulations of the proposed system were performed with Hybrid Optimization of Multiple Energy Resources (HOMER) software. In order to determine the optimal combination, the wind and PV system as well as the batteries and generator systems were selected on the basis of the Net Present Cost and the highest renewable energy production from the simulation results. The general hospitals load consists of 430kW peak and 6MWh/d. The results show optimization performance for production at PV array 1507.73MWh/yr, wind turbine 596.81MWh/yr and generator 588.31 MWh/yr while the cost of energy was found to be \$0.182/KWh.

Keywords—hybrid, Design, renewable, optimization. Simulation

I. INTRODUCTION (RENEWABLE ENERGIES)

The increased industrialization and a rising world population led to the growth in electricity demand. Traditional electricity generation systems use non-renewable resources such as coals, gas, and oil. Nevertheless, such resources contribute to the rise in the emission of greenhouse gasses (GHGs) and carbon dioxide (CO₂). GHGs and CO₂ are known as hazardous to our environment and the atmosphere [1]. Among adverse consequences of fossil fuel consumption are deterioration, biodiversity loss, deforestation, coastal pollution, and air pollution [1]. Hence, environmentalists, policymakers, and researchers are all keenly interested in exploring potential renewable energy resources to come at the place the conventional fossil fuel and to reduce the emission [2]. Among renewable energy sources, there are different forms of biomass, hydropower, solar, geothermal, and resources. Apart from hydropower, the most widely used are solar and wind resources [3]. The reason is the easy and feasible conversion of solar and wind energy into electricity, which is not the case of wave and tide

energy [3]. Hence, it makes the production of electricity from these renewable resources very economically [4].

II. PROSPECTS OF RENEWABLE ENERGY SOURCES

Fossil fuels such as coal and oil are not sustainable resources, as they become depleted when used and take a long time to regenerate [5][6]. However, Renewable energies, unlike hydrocarbon energies, are developed from constantly replenished resources that don't deplete [6].

These above-listed renewable energy sources are clean and 'green' (i.e., environmentally friendly) sources of energy and have minimal environmental impacts. Unlike fossil fuels, renewable resources are fairly evenly distributed all over the world and are available at no or minimal cost. Only a small fraction (about 12%) of the world's primary energy is provided via renewable energy sources (MATHEMA, 2013). Ever-increasing energy needs, and depleting fossil fuel reserves have led to the increasing use of renewable energy sources to satisfy energy demands. Globally, 1.4 billion people live without electricity and many people around the world face recurrent power outages. The subsequent growth in electricity demands, together with concerns related to climate change and the practical suitability of other technologies, has stirred interest in renewable energy technologies [6]. Efficient use of diverse renewable energy sources can pave the way for sustainable development. Therefore, careful design and planning for the utilization of renewable energy sources is required for the sustainable development of a society. Renewable energy sources are capable of addressing global problems of energy security, climate change and sustainable development, so there is a pressing need to accelerate the development of renewable energy technologies [7].

III. HYBRID ENERGY SYSTEMS

A system that combines more than one energy source or uses different conversion devices is known as a hybrid energy system (Celik, 2002). Hybrid energy systems supply the generated electricity either directly to the grid (Yang, Lu, & Burnett, 2003) or to a battery bank for storage (Shaahid & Elhadidy, 2003). Therefore, a diesel generator is mainly used as a backup. There are three different configurations for hybrid systems which are grid connected, off grid with distribution systems and direct supply [7]. Grid connected

configuration can rely on the grid to cover any deficiency in meeting the load demand. The hybrid systems can either be connected to many load centers or can act as a source of supply for one or two loads, thus avoiding the need of a distribution system in off-grid configurations. An isolated off-grid system is usually used to charge batteries or supply power to small rural industry/households (Rakesh, Digbijoy, Tanushree, & Nishant, 2016). National Renewable Energy Laboratory (NREL)'s, Hybrid Optimization Model for Electric Renewable (HOMER version 2.19) has been used as the sizing and optimization software too(National Renewable Energy & Laboratory, 2016).

IV. SITE INFORMATION OF CITY OF SABRATHA

The site chosen for this project is in Libya of Sabratha. As mentioned before the goal of this project is to power Sabratha's general hospital with an area of 9000m² that is built on government owned land. The site is situated near the sea which decreases the cost of cabling when using offshore turbines, the site is not surrounded by any tall buildings that could block the wind or the sun. The site is near the coastal road (1.31Km) which makes it easier for transporting equipment. The building's roof is also large enough to contain the chosen PV system.

V. ENERGY RESOURCES IN SABRATHA (LIBYA)

Due to its geo-climatic conditions, Sabratha is blessed with several forms of renewable energy resources which includes solar, wind and tidal power. However, this project will only focus on solar and wind power.

A. Wind characteristics in Sabratha

The annual average wind speed in Sabratha according to NASA (National Aeronautics and Space Administration) is 5 m/s and mostly blows from the ENE (east north east) direction as shown in figure (1) (Windfinder, 2018). The highest average monthly wind speeds occur in November, December, January, and February and the lowest speeds occur during June as shown in figure (1). These statistics are based on observations taken between 11/2007 – 03/2018

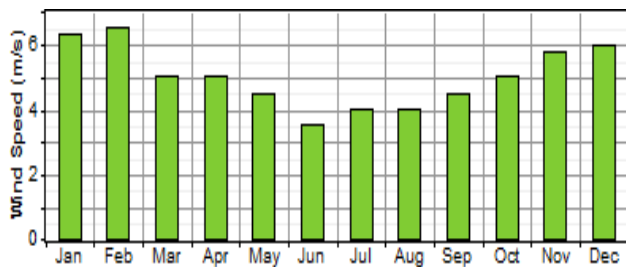


Fig. 1. Wind resource of Sabratha in Libya in HOMER

B. Solar characteristics in Sabratha

The annual average solar irradiance in Sabratha according to NASA is 5.5kWh/m²/day with 5.5 hours of peak sunlight. June and July had the highest solar irradiance and the highest clearness index while January and December had the lowest irradiance and clearness index as shown in figure (2) [8].

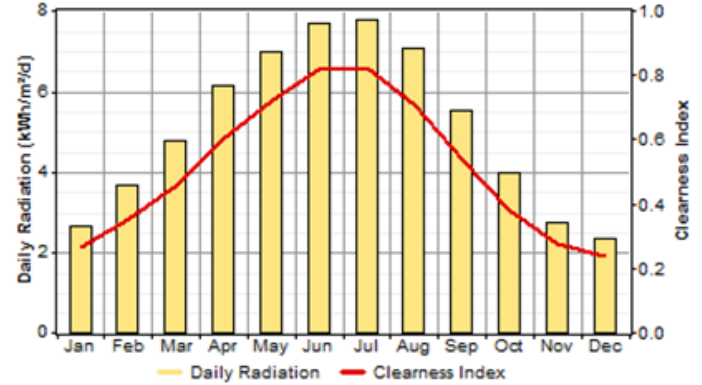


Fig. 2. Monthly solar radiation using the HOMER software tool chart based on data from NASA.

C. calculation Load of sabratha city

To calculate the building load two methods were chosen, the first method was to gain data from the general electricity company and the second was to go to the site and calculate the load manually. The first method, according to an employee at the electric company, was found to be inaccurate therefore the second method was adopted. The second method included calculating the daily load for each of the building's electrical appliances e.g., lighting, medical equipment, air conditioners, etc. The load calculation is shown in table (I) and figure (3), where equation (1) was used to calculate the total daily load.

$$\text{Daily load} = \sum (\text{application power} \times \text{working hours} \times \text{no. application}) \quad (1)$$

TABLE I SHOWN CALCATION OF LOAD

Appliance	Power (W)	Working hours (h)	Number	Consumption per day (Wh/day)
Lamps	18	15	8000	2160000
Computers	350	6	30	63000
Air conditioners	1800	8	200	2880000
Boiler	1000	5	70	350000
TV	120	12	30	43200
Fridge	150	16	15	36000
C.T scanner	2000	4	1	8000
Incubators	350	24	12	100800

X-ray	3000	15	2	90000
Ultrasound	1350	12	2	32400
Elevator	7460	15	2	223800
		Working hours (h)	5987200	
		Total Daily Consumption (MW)	5.9872	

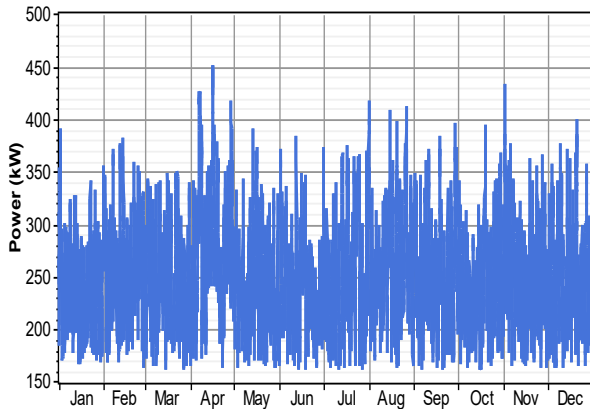


Fig. 3. Fig. a1 load profile of hosbatal in Sabratha

VI. DESIGN OF SYSTEM

A. Design of wind turbine

The chosen wind turbine for the wind system is a ReDriven 10KW onshore turbine as to utilize the power output of lower wind speeds since they operate at low cut in speeds. The wind turbine will face the east north east direction i.e., 67.5° from the north since, as mentioned before, the wind mostly blows from that direction.

Kinetic energy exists whenever an object of a given mass is in motion with a translational or rotational speed. When air is in motion, the kinetic energy in moving air can be determined as:

$$K_E = \frac{1}{2} \times m \times V^2 \quad (2)$$

The wind power can be obtained by differentiating the kinetic energy in wind with respect to time, Equation (3) was used to calculate the power produced by the wind.

$$P_{Wind} = \frac{1}{2} \times V^3 \times \rho \times A \quad (3)$$

Table (II) shows the turbines PAero of each wind speed while figure (4) shows the turbines power curve.

TABLE II SHOWS THE TURBINES PAERO OF EACH WIND SPEED

Wind speed (m/s)	PAero(kW)	Wind speed (m/s)	PAero (kW)
1	0	11	10
2	0.11088	12	10
3	0.37422	13	10
4	0.88704	14	10
5	1.7325	15	10
6	2.99376	16	10
7	4.75398	17	10
8	7.09632	18	10
9	10	19	0
10	10	20	0

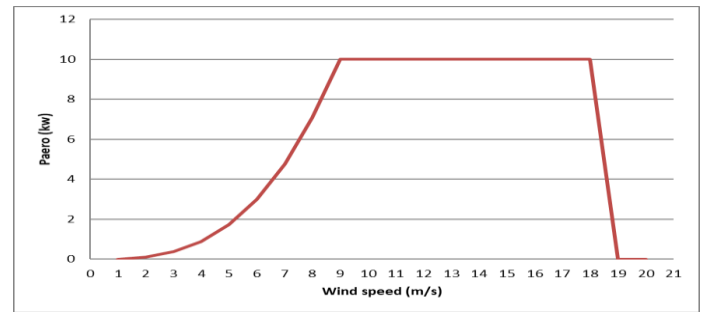


Fig. 4. Wind turbine power curve

To calculate the probability of each wind speed occurring equation (4) was used while equation (5) was used to calculate the annual hours of each wind speed.

$$\text{wind speed}\% = \frac{\pi}{2} \frac{v}{\bar{v}^2} \exp \left[-\frac{\pi}{4} \left(\frac{v}{\bar{v}} \right)^2 \right] \quad (4)$$

$$\text{Annual wind hrs} = \text{wind speed}\% \times 24 \times 365 \quad (5)$$

Where wind speed% represents probability of each wind speed. figure (5) shows the annual hours per wind speed.

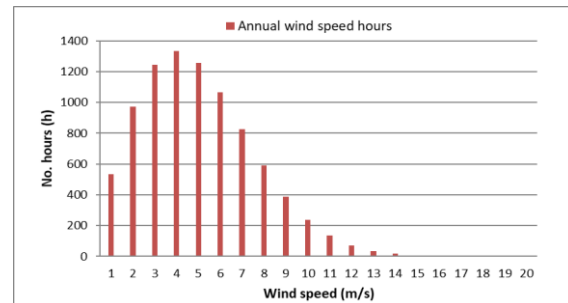


Fig. 5. the annual hours per wind speed

B. Design of solar panel

For the solar system, a 350w poly-crystalline PV array made by SunPower® was chosen to maximize the production while minimizing the required area at a reasonable price (Sunpower Corporation, 2016). Table (III) shows the characteristics of the chosen PV panel and figure ((6) shows the dimensions of the panel.

TABLE III SHOWS THE CHARACTERISTICS OF THE PV PANEL

Nominal Power (PNOM)	350 W
Price	400\$
Rated Voltage (VMPP)	43.1 V
Rated Current (IMPP)	8.12 A
Open-Circuit Voltage (VOC)	51.7 V
Short-Circuit Current (ISC)	8.65 A
Life time	20years

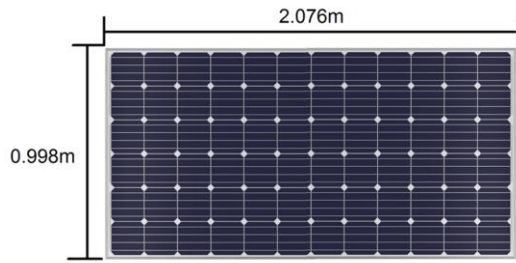


Fig. 6.the dimensions of the panel

To calculate the efficiency of the panel equation (6) is used, it should be mentioned that the maximum solar power radiation reaching the earth surface is 1000W/m² (Group, 2013).

$$\text{olar cell efficiency (\%)} = \frac{\text{Panel Nominal Power}}{\text{Panel Area} \times \text{Maximum solar power}} \times 100 \quad (6)$$

The solar panels efficiency was found to be 16.97% which is an acceptable efficiency since commercial solar panels have an efficiency range between 15-21% (EnergySage, 2018). the Capacity of the PV system. Equation (7) is used to calculate the peak sun hours while equation (8) is used to calculate the Capacity of the PV panels [9].

$$\text{Peak Sun Hours} = \frac{\text{Avg. Daily Irradiation}}{\text{Peak Sun Irrerdance}} \quad (7)$$

$$\text{Capacity of the PV panels} = \frac{\text{Daily load covered} \times 1.3}{\text{Peak Sun Hours}} \quad (8)$$

The peak sun hours were found to be 5.5 hours and the capacity of the PV system was calculated to be 945.45kW with a daily consumption of 4MWh. thus, can be calculate the number of solar panels needed equation (9) is used.

$$\text{No. of solar panels} = \frac{\text{Capacity of the PV system}}{\text{Capacity of each panel}} \quad (9)$$

The number of solar panels needed was found to be 2702 panels how ever before calculating the required area it is necessary to study the effect of shading.

VII. SIMULATION RESULTS VIA HOMAER SOFTWARE

A. Hybrid system schematic diagram

The diagram for the first case used in HOMER is shown in figure (7), which consists of a PV system, wind system, battery bank (S4KS25P), inverter (converter), generator and a daily primary load of 6MW.

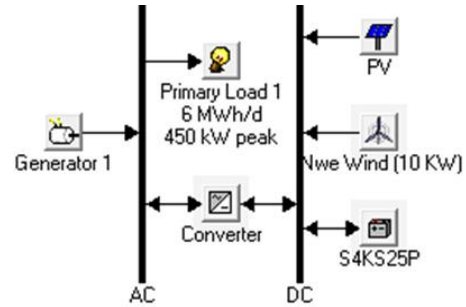


Fig. 7.Hybrid system schematic diagram for technical institutes

In order to ensure that the chosen system configuration is the most optimum, different size for the solar and wind system will be taken into consideration. The PV capacities within the range of 50-950 kW will be used while the number of wind turbine considered will be in the range of 1-23 turbines. choose the most optimum configuration based on the levelized cost of electricity (the cost of producing a kW of energy) and present the most viable options. Figure (8) shows the most viable system configurations which are sorted in descending based on the levelized cost of electricity where the first configuration is the most optimum.

	PV (kW)	G10	Gener (kW)	S4KS25P	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Diesel (L)	Gener (hrs)
1	950	23	350	1728	500	\$ 1,976,917	213,532	\$ 4,706,569	0.168	0.81	182,981	1,818
2	950	22	300	576	500	\$ 420,400	483,910	\$ 6,606,397	0.236	0.28	576,386	6,077
3	950		350	2016	500	\$ 1,977,117	368,357	\$ 6,685,955	0.239	0.58	372,777	3,471
4	850	23	400		500	\$ 1,176,431	456,591	\$ 7,013,193	0.251	0.63	529,457	6,851
5		23	400		250	\$ 180,000	580,268	\$ 7,597,767	0.271	0.29	707,514	8,760
6	950		400		500	\$ 1,175,717	508,321	\$ 7,673,764	0.274	0.51	596,877	6,948
7			350	288	250	\$ 175,200	649,717	\$ 8,480,770	0.303	0.00	793,782	8,760
8			450		250	\$ 70,000	700,961	\$ 9,030,631	0.323	0.00	862,859	8,760

Fig. 8.Optimization results of hybrid energy system

B. Cost summary

The cost summary for each system component includes the sum of capital, replacement, operating and fuel cost while subtracting the salvage cost. Table (IV) shows a detailed explanation of the cash flow summary while, figure (9) displays the cash flow summary.

TABLE IV SHOWS A DETAILED EXPLANATION OF THE CASH FLOW SUMMARY

System Component	Capital cost (\$)	Replacement cost (\$)	Operating and maintenance cost (\$)	Fuel cost (\$)
Pv	1,085,717	338,532	138,687	0
Wind	115,000	35,858	44,103	0
Generator	35,000	0	16,268	1,871,285
Battery	691,200	514,217	110,448	0
Converter	50,000	15,590	12,783	0
Overall system	1,976,917	904,197	322,289	1,871,285

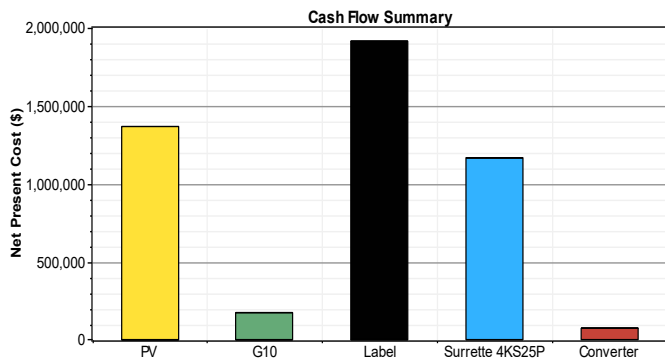


Fig. 9. Displays the cash flow summary

As it can be seen PV system had the highest costs excluding fuel costs which is only relevant for the generator. The overall system total cost was \$4,706,572 with the capital cost of the PV system and the fuel cost of the generator being the greatest contributors to the cost (approximately 62.83%).

C. Electric Production

The production monthly average electric production for each system component is shown in figure (10) where the PV system is represented by yellow, the wind system is represented by green and the generator by black.

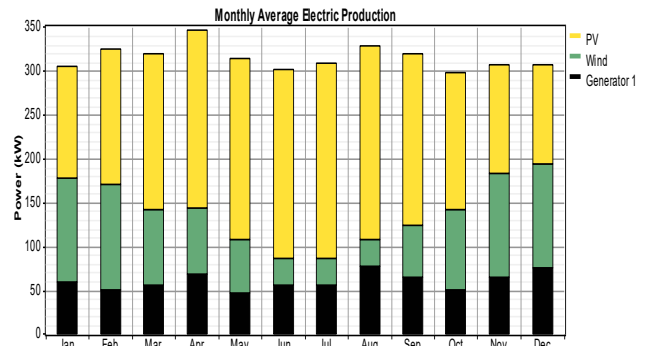


Fig. 10. Monthly average electricity production

The previous figure the proposed system was able to cover the daily average load of 250 kW where most of the load is covered by the PV system especially in summer while during the winter the solar and wind production were very similar. As for the generator it had a similar output throughout the year which increases or decreases in order to cover any unmet load. Table (V) shows the annual production of each system.

TABLE V SHOWS THE ANNUAL PRODUCTION OF EACH SYSTEM

Type	Production (kWh/yr)	%
PV array	1,545,142	56
Wind turbines	685,908	25
Generator1	528,306	19
Total	2,759,356	100

As it can be seen from the previous table, the PV arrays covered, on average, 56% while the wind covered 25%. The percentage of the load covered by each resource depends heavily on the resources of the site and can vary from one year to another. As for the generator, it was set to cover any unmet load which corresponds to 19% of the time therefore the proposed hybrid system has an 81% renewability factor since the generator does not constitute a renewable source.

VIII. CONCLUSIONS

The paper presents a Modelling and simulation of a hybrid power system useful for the based on renewable energy. The hybrid power system was proposed as an ultra-low emission energy system. The simulation results indicated that a system comprised of a load of 6 MW peak and 450 KW peak and a system supply of two panels provides 350 watts at Rated Voltage (VMPP) 43.1V with a The maximum output was obviously 230kW since the rated power of a single turbine was 10kW however the average output of the system was only 78kW which is only 33.91% of the maximum output,

and The battery bank received annually 1106679kWh of energy and provided 891805kWh with 7277kWh of energy lost due to storage depletion and 207597kWh lost due to other losses which corresponds to 19.41% loss. The presented simulation model also represents a useful tool in energy management system domain. Therefore, for future developments and facilities is considered the opportunity of current and voltage.

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